ECEN 662: Estimation & Detection Theory

Lecture 18: Monte Carlo Integration and Importance Sampling

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- Consider the generic problem of evaluating the expectation/integral

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x) f(x) dx$$

- For cases where the above expectation/integral *cannot* be evaluated analytically, it is natural to propose using multiple *independent* samples $(X_1, \ldots, X_m)$ drawn from the density f to approximate it:

$$\hat{h}_m = \frac{1}{m} \sum_{i=1}^{m} h(X_i)$$

- This approach is often referred to as the *Monte Carlo* method

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- In practice, however, the sample size m is always *finite*. In order for the method to work, the *variance* of the estimate

$$\mathrm{Var}_f[\hat{h}_m] = \frac{\mathrm{Var}_f[h(X)]}{m}$$

has to be *sufficiently small*

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- In this case, $h(X) \sim \mathcal{B}(p_B)$, where

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so

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- If we require the standard deviation of $\hat{h}_m$ to be in the *same order* as p_B, we need

$$\sqrt{\frac{p_B - p_B^2}{m}} \approx p_B \quad \Longrightarrow \quad m \approx \frac{1}{p_B} - 1$$

which can be exceedingly *large* for a very *rare* event B

Importance sampling

- The method of *importance sampling* is an evaluation of the expectation/integral

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- The method is based on the following alternative representation of the expectation/integral:

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x)\omega(x)g(x)dx = \mathbb{E}_g\left[\omega(X)h(X)\right]$$

where $\omega(x) = f(x)/g(x)$ is the density ratio

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- The above representation immediately leads to the following approximation based on samples $(X_1, \ldots, X_m)$ drawn from g:

$$\hat{h}_m = \frac{1}{m}\sum_{i=1}^{m} \omega(X_i)h(X_i)$$

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- Importance sampling is thus of considerable interest because it puts very *little* restriction on the choice of the proposal distribution, which can be chosen from distributions that are easy to sample from
- Moreover, the same samples from g can be used *repeatedly* for different functions h but also for different densities f

Variance control

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- Among proposal distributions leading to finite variances for the importance sampling estimator, it is possible to identify *optimal* proposal distribution corresponding to a given function h and a fixed distribution f

- More specifically, the choice of g that minimizes the *variance* of the importance sampling estimator is

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- From a practical point of view, the above result suggests looking for proposal distributions g for which $\omega(x)|h(x)|$ is almost *constant*

Self-normalizing importance sampling

- An alternative to the vanilla importance sampling estimator

$$\hat{h}_m = \frac{1}{m} \sum_{i=1}^{m} \omega(X_i) h(X_i)$$

is the following *self-normalizing importance sampling* estimator

$$\hat{h}'_m = \frac{\sum_{i=1}^{m} \omega(X_i) h(X_i)}{\sum_{i=1}^{m} \omega(X_i)}$$

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- Note that as $m \to \infty$,

$$\frac{1}{m}\sum_{i=1}^{m}\omega(X_i) \overset{i.p.}{\sim} \mathbb{E}_g[\omega(X)] = \int_{\mathcal{X}} \frac{f(x)}{g(x)} g(x)dx = \int_{\mathcal{X}} f(x)dx = 1$$

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- Therefore, just like the vanilla importance sampling estimator, the *self-normalizing* importance sampling estimator is also a *consistent* estimator of the expectation $\mathbb{E}_f[h(X)]$

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- In addition, the *self-normalizing* importance sampling estimator also enjoys a couple of important additional benefits
- First, unlike the vanilla importance sampling estimator, the *self-normalizing* importance sampling estimator is slightly *biased* and thus may enjoy the benefit of having a *smaller* variance
- Second, unlike the vanilla importance sampling estimator, which requires precise knowledge of the density f, the *self-normalizing* importance sampling estimator can be implemented based on an *un-normalized* density $\tilde{f}$ with an *unknown* scaling factor, which makes it particularly appealing when f is a *posterior* distribution in Bayesian statistics

Monte Carlo marginalization

- Consider the problem of computing the marginal density $f(x)$ from the joint density $f(x, z)$ via *importance sampling*:

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- Based on the previous analysis, the *optimal* proposal distribution that minimizes the variance must ensure that the ratio $f(x, z)/g(z)$ is *constant* with respect to z:

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  and hence

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- However, knowing the joint density $f(x, z)$, computing the posterior distribution $f(z|x)$ is the *same* as computing the marginal distribution $f(x)$

- Therefore, to perform *marginalization* using importance sampling, one may wish to find an *easy-to-sample-from* and *easy-to-evaluate* proposal distribution that also *approximates* the posterior distribution

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- This is the essential idea of *variational auto-encoding (VAE)*

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- The conditional distribution $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ must also be *easy* to draw from
- The generative model $p_\theta(\boldsymbol{x})$ is *implicitly* defined as the *marginal* distribution of $\boldsymbol{x}$:

$$p_\theta(\boldsymbol{x}) = \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} = \mathbb{E}_{\boldsymbol{Z} \sim p(\boldsymbol{z})}[p_\theta(\boldsymbol{x}|\boldsymbol{Z})]$$

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- In statistics, one often use the so-called *K-L divergence* to measure the "distance" between two probability distributions p and q:

$$D_{KL}(p\|q) := \int p(x) \log \frac{p(x)}{q(x)} dx = \mathbb{E}_{X\sim p(x)} \left[\log \frac{p(X)}{q(X)} \right]$$

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- One possibility is to learn a parameter θ by minimizing the K-L divergence

$$\begin{aligned} D_{KL}(p\|p_\theta) &= \mathbb{E}_{\boldsymbol{X}\sim p(\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{X})}{p_\theta(\boldsymbol{X})} \right] \\ &= \mathbb{E}_{\boldsymbol{X}\sim p(\boldsymbol{x})} \left[\log p(\boldsymbol{X}) \right] - \mathbb{E}_{\boldsymbol{X}\sim p(\boldsymbol{x})} \left[\log p_\theta(\boldsymbol{X}) \right] \end{aligned}$$

Maximum-likelihood (ML) estimate

- Note that minimizing the K-L divergence $D_{KL}(p\|p_\theta)$ with respect to θ is equivalent to maximizing

$$\mathbb{E}_{\boldsymbol{X}\sim p(\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{X})\right] \approx \frac{1}{m}\sum_{i=1}^{m}\log p_\theta(\boldsymbol{x}_i)$$

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- To compute the maximum-likelihood estimate, we need to evaluate the marginal distribution

$$p_\theta(\boldsymbol{x}) = \int p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})d\boldsymbol{z} = \mathbb{E}_{\boldsymbol{Z}\sim p(\boldsymbol{z})}[p_\theta(\boldsymbol{x}|\boldsymbol{Z})]$$

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- Note, however, that the integral for evaluating $p_\theta(\boldsymbol{x})$ *cannot* be evaluated analytically

Monte-Carlo (MC) integration

- In practice, one may *estimate* $p_\theta(\boldsymbol{x})$ through *Monte Carlo integration*:

$$p_\theta(\boldsymbol{x}) \approx \frac{1}{n} \sum_{j=1}^{n} p_\theta(\boldsymbol{x}|\boldsymbol{z}_j)$$

  where $(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n)$ are independent samples of the random source $\boldsymbol{z}$

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- However, the conditional probability $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ usually *fluctuates* significantly for different values of $\boldsymbol{z}$. As a result, the above estimate only becomes reliable when the number of samples drawn from the random source $\boldsymbol{z}$ becomes very large

Importance sampling to the rescue

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- The main idea is to introduce a new *sampling distribution* $q(\boldsymbol{z})$ to rewrite the expression for $p_\theta(\boldsymbol{x})$:

$$\begin{aligned} p_\theta(\boldsymbol{x}) &= \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} \\ &= \int \frac{p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z})}{q(\boldsymbol{z})} q(\boldsymbol{z}) d\boldsymbol{z} \\ &= \mathbb{E}_{\boldsymbol{Z} \sim q(\boldsymbol{z})} \left[\frac{p(\boldsymbol{Z}) p_\theta(\boldsymbol{x}|\boldsymbol{Z})}{q(\boldsymbol{Z})} \right] \\ &\approx \frac{1}{n} \sum_{j=1}^{n} \frac{p(\boldsymbol{z}_j) p_\theta(\boldsymbol{x}|\boldsymbol{z}_j)}{q(\boldsymbol{z}_j)} \end{aligned}$$

where $(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n)$ are independent samples drawn according to $q(\boldsymbol{z})$

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- Letting

$$\frac{p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})}{q(\boldsymbol{z})} = c$$

  gives

$$q(\boldsymbol{z}) = \frac{p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})}{c}$$

  where the constant c needs to be chosen such that

$$\int q(\boldsymbol{z})d\boldsymbol{z} = 1$$

- This gives

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- Unfortunately, the posterior distribution $p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is *untenable* because evaluating it requires the knowledge of the marginal distribution $p_\theta(\boldsymbol{x})$
- In practice, one may introduce a second parameterized model $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ to *approximate* the posterior distribution $p_\theta(\boldsymbol{z}|\boldsymbol{x})$

The encoder and decoder

- To summarize, even though our original goal was simply to train a generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ via maximum-likelihood estimate, estimating the marginal likelihood $p_\theta(\boldsymbol{x})$ requires training an additional sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x}) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$

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- Further note that while the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ *generates* data from random source, the goal of the sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x}) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is to *recover* the random source from the data

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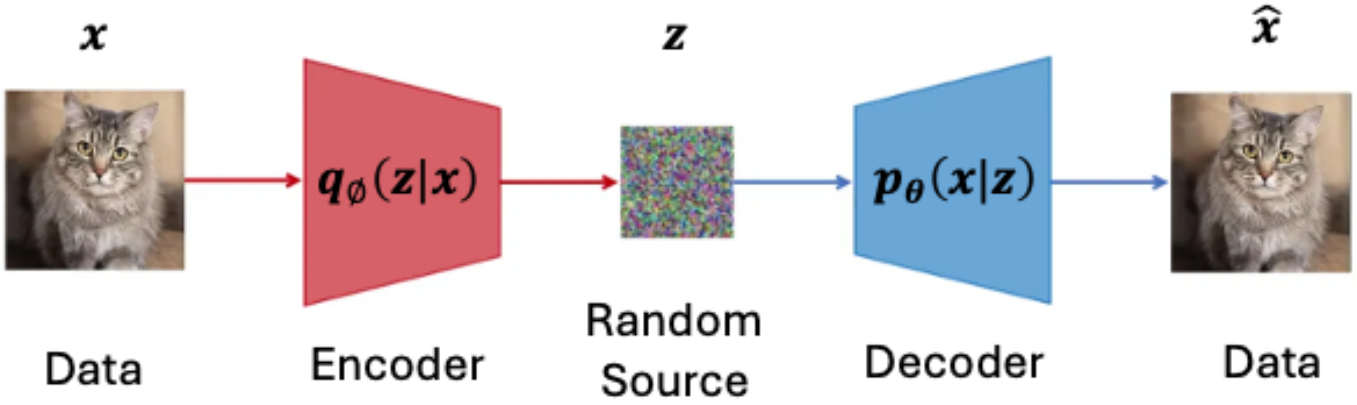
- Further note that while the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ *generates* data from random source, the goal of the sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x}) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is to *recover* the random source from the data
- We can think of the role of generator and sampling distribution as a process of first transforming data into a random source using $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ and then recovering the data using the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$



- This process of *data self-recovery* is usually known as *auto-encoding*. Under the context of *auto-encoding*, we call the sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ the *encoder* and the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ the *decoder*

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- It is important to note that the aforementioned auto-encoding is *stochastic* in nature in that the recovered data is only equivalent to the original data *statistically* and in general is *not* the same as the original data

Joint training of encoder and decoder

- To jointly learn the encoder and decoder, let us consider the following *variational* representation of $\log p_\theta(\boldsymbol{x})$:

$$\log p_\theta(\boldsymbol{x}) = \max_{q(\boldsymbol{z}|\boldsymbol{x})} \left[\log p_\theta(\boldsymbol{x}) - D_{KL}(q(\boldsymbol{z}|\boldsymbol{x})||p_\theta(\boldsymbol{z}|\boldsymbol{x}))\right]$$

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- The above representation holds due to the fact that the K-L divergence

$$D_{KL}(q(\boldsymbol{z}|\boldsymbol{x})\|p_\theta(\boldsymbol{z}|\boldsymbol{x})) \geq 0$$

with the equality holds if and only if

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- Note that for any given $\boldsymbol{x}$,

A variational characterization of the log likelihood

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\begin{aligned}
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Modeling the sampling distribution

- To use *Monte Carlo* sampling to estimate the objective function

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- We thus introduce a second parameterized *model* $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ and use it to give an *approximate* expression for $\log p_\theta(\boldsymbol{x})$:

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- Note that in order for the above approximation to be accurate, the model class needs to be sufficiently large so that one can always find a sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ such that

$$q_\phi(\cdot|\boldsymbol{x}) \approx p_\theta(\cdot|\boldsymbol{x})$$

Importance sampling at work

- The key point here is that when $q_\phi(\cdot|\boldsymbol{x}) \approx p_\theta(\cdot|\boldsymbol{x})$, it is *easier* to use Monte Carlo integration to estimate the objective function

$$\mathbb{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{Z})p_\theta(\boldsymbol{x}|\boldsymbol{Z})}{q_\phi(\boldsymbol{Z}|\boldsymbol{x})}\right]$$

because the expectant

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- Conclusion: We can *jointly* learn an encoder and a decoder by maximizing

$$\frac{1}{m}\sum_{i=1}^{m} \mathbb{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x}_i)}\left[\log \frac{p(\boldsymbol{Z})p_\theta(\boldsymbol{x}_i|\boldsymbol{Z})}{q_\phi(\boldsymbol{Z}|\boldsymbol{x}_i)}\right]$$

  over both θ and ϕ, where the expectation over $\boldsymbol{Z}$ can be estimated using Monte Carlo integration

Rejection sampling *v.s.* importance sampling

- Finally, let us make an informal comparison between rejection sampling and importance sampling in terms of computing the expectation/integration

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x)f(x)dx$$

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- As discussed before, the *self-normalizing* importance sampling estimator is given by

$$\hat{h}'_m = \frac{\sum_{i=1}^m \omega(X_i)h(X_i)}{\sum_{i=1}^m \omega(X_i)}$$

where $(X_1, \ldots, X_m)$ are independent samples drawn from the proposal distribution g

- Alternatively, one may select a *subset* of samples $(X_{i_1}, \ldots, X_{i_k})$ from $(X_1, \ldots, X_m)$ using the *acceptance* rule

$$U_i \leq \frac{\omega(X_i)}{M}$$

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- The sub-samples $(X_{i_1}, \ldots, X_{i_k})$ are independent samples from f and they can be used to perform classical Monte Carlo integration:

$$\hat{h}''_m = \frac{1}{k} \sum_{j=1}^{k} h(X_{i_j}) = \frac{\sum_{i=1}^{m} \tilde{\omega}(X_i, U_i) h(X_i)}{\sum_{i=1}^{m} \tilde{\omega}(X_i, U_i)}$$

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- Note the *similarity* between the above two estimators